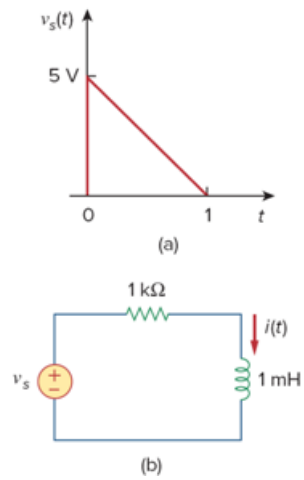


Problem

Given the circuit in Fig. 18.41, with its excitation, determine the Fourier transform of $i(t)$.

Figure 18.41



Step-by-step solution

Step 1 of 3

Refer to Figure 18.41(a) in the textbook for input waveform.

From the waveform, the voltage signal can be expressed as,

$$v_s(t) = \begin{cases} -5t + 5 & \text{for } 0 \leq t \leq 1 \\ 0 & \text{other wise} \end{cases}$$

Apply Fourier transform to the voltage signal.

$$\begin{aligned} V_s(\omega) &= \int_{-\infty}^{\infty} v_s(t) e^{-j\omega t} dt \\ &= \int_0^1 (-5t + 5) e^{-j\omega t} dt \\ &= -5 \int_0^1 t e^{-j\omega t} dt + 5 \int_0^1 e^{-j\omega t} dt \\ &= -5 \left[\frac{t e^{-j\omega t}}{-j\omega} \right]_0^1 - \int_0^1 \left(\frac{e^{-j\omega t}}{-j\omega} \right) dt + 5 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 \\ &= -5 \left[\frac{1 e^{-j\omega 1} - 0 e^0}{-j\omega} - \left[\frac{e^{-j\omega t}}{(-j\omega)^2} \right]_0^1 \right] + 5 \left[\frac{e^{-j\omega t}}{-j\omega} \right]_0^1 \\ &= -5 \left[\frac{e^{-j\omega} - 0}{-j\omega} - \left[\frac{e^{-j\omega 1} - e^0}{(-j\omega)^2} \right] \right] + 5 \left[\frac{e^{-j\omega 1} - e^0}{-j\omega} \right] \\ &= -5 \left[\frac{e^{-j\omega}}{-j\omega} - \frac{e^{-j\omega 1} - 1}{(-j\omega)^2} \right] + 5 \left[\frac{e^{-j\omega 1} - 1}{-j\omega} \right] \\ &= \left[\frac{-5e^{-j\omega} + 5e^{-j\omega} - 5}{-j\omega} + \frac{5e^{-j\omega} - 5}{(-j\omega)^2} \right] \\ &= \left[\frac{0 - 5}{-j\omega} - \frac{5e^{-j\omega} - 5}{\omega^2} \right] \\ &= \left[\frac{5}{j\omega} - \frac{5e^{-j\omega} - 5}{\omega^2} \right] \end{aligned}$$

Step 2 of 3

Refer to Figure 18.41(b) in the textbook.

Draw the circuit diagram in frequency domain.

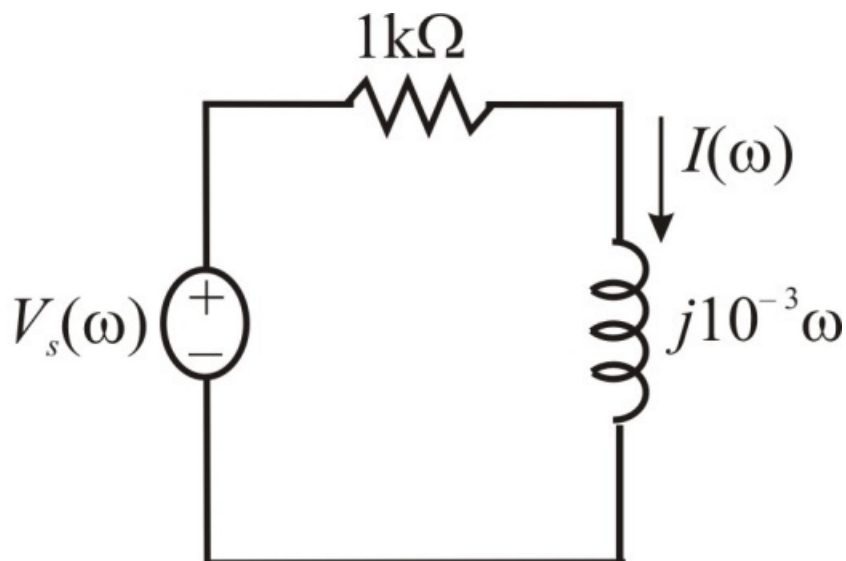


Figure 1

Step 3 of 3

From Figure 1, the current in the circuit is,

$$I(\omega) = \frac{V_s(\omega)}{10^3 + j10^{-3}\omega}$$

Substitute $\frac{5}{j\omega} - \frac{5e^{-j\omega} - 5}{\omega^2}$ for $V_s(\omega)$.

$$\begin{aligned} I(\omega) &= \frac{\frac{5}{j\omega} - \frac{5e^{-j\omega} - 5}{\omega^2}}{10^3 + j10^{-3}\omega} \\ &= \frac{5 \times 10^3}{10^6 + j\omega} \left(\frac{1}{j\omega} - \frac{e^{-j\omega} - 1}{\omega^2} \right) \\ &= \frac{5 \times 10^3}{10^6 + j\omega} \left(\frac{1}{j\omega} + \frac{1}{\omega^2} - \frac{e^{-j\omega}}{\omega^2} \right) \end{aligned}$$

Thus, the Fourier transform of $i(t)$ is $I(\omega) = \frac{5 \times 10^3}{10^6 + j\omega} \left(\frac{1}{j\omega} + \frac{1}{\omega^2} - \frac{1}{\omega^2} e^{-j\omega} \right)$.